

CoinRoutes Frontend Code Test

Before beginning, read about how an order book works here

<https://falconair.github.io/2015/01/05/financial-exchange.html> (you can ignore the scala bits)

Since they use slightly non-standard terminology, here are some definitions of terms this description uses:

- **Bids:** the line of buyers as referenced in the article.
 - Array of records. Each record contains Quantity and price. Sorted by highest price first.
- **Offers:** The line of sellers as referenced in the article.
 - Array of records. Each record contains Quantity and price. Sorted by lowest price first.
- **Order Book:** This is an object that contains the bids and offers data arrays.

The goal of this code test is to test your ability to create a simple interactive web-application using **React** that can consume real-time data and display it to the user in a usable way. Please use functional components and hooks. Components should be composable and reusable when appropriate. You can use Redux or another state management library, however it is not required.

You will be consuming websocket order-book data from an exchange, and displaying a few widgets with this data. You are free to use third-party charting / drawing libraries as you see fit, however you will be evaluated on the modularity and efficiency of your solution

Expected output:

User inputs:

The UI should consist of a Drop-down where the user can select a number of currency pairs. For the purposes of this example the currencies should be: BTC-USD, ETH-USD, LTC-USD, BCH-USD. These widgets should be rendered on screen for each currency pair selected.

Data sources

You should use the Coinbase Pro order book data feed:

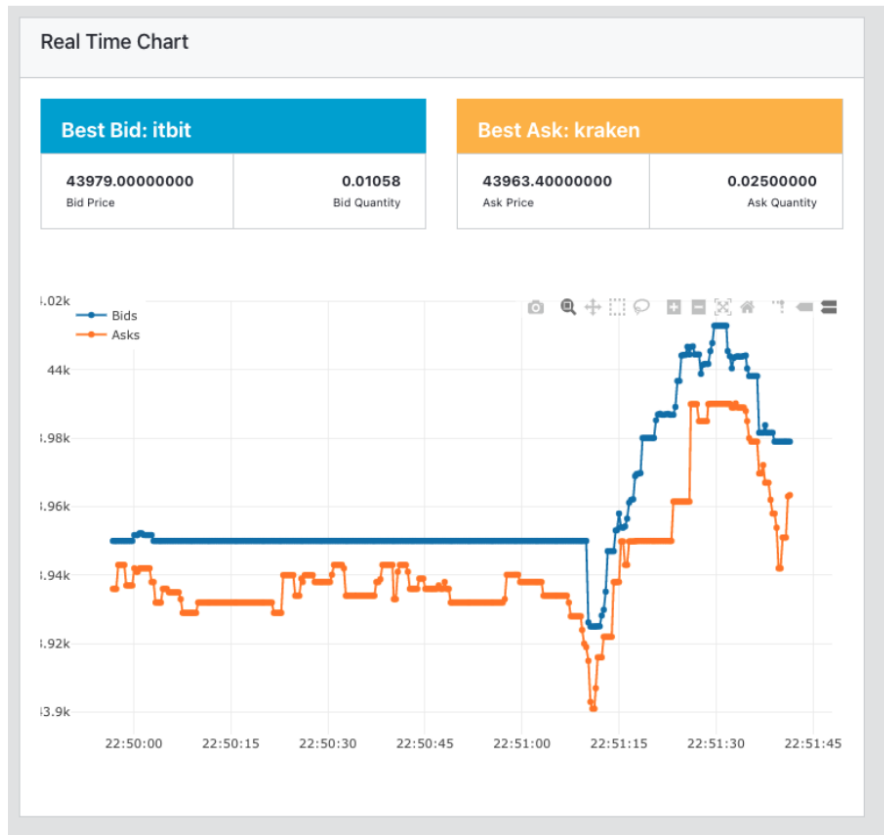
<https://docs.cloud.coinbase.com/exchange/docs/websocket-channels#level2-channel>

(continued below...)

Data display:

There should be a few different widgets :

- Top of book, IE display the best bid and ask of the selected currency pair
- Price Chart. Use a charting library to display the current price chart



- "Ladder" view: show a ladder view of the data. Note that in crypto there are many more price levels than in other asset classes, so as a bonus allow aggregating the prices by set price increments.

Looking at the picture *below* (next page), you can see that it would allow you to change the increment from 0.01 to 0.05, 0.10, etc... If the aggregation was \$0.10, you would add up the quantity of each increment and only display one price every \$0.10

Order Book			
	Market Size	Price (USD)	My Size
	0.0329	44202.80	-
	0.0500	44200.00	-
	0.0984	44198.97	-
	0.4527	44198.94	-
	0.4000	44198.93	-
	0.0342	44195.55	-
	0.5000	44195.54	-
	0.4883	44194.26	-
	0.0347	44192.64	-
	0.0472	44191.77	-
	0.9031	44191.19	-
	0.2263	44190.90	-
	0.0500	44190.80	-
	0.0059	44188.01	-
	0.0821	44187.93	-
	0.2250	44187.34	-
	0.0165	44183.36	-
	0.0066	44183.31	-
	0.2250	44183.26	-
	0.2250	44180.46	-
	0.0054	44180.02	-
	14.8227	44179.00	-
	2.8541	44173.91	-
USD Spread		0.01	
	3.6601	44173.90	-
	0.0400	44173.86	-
	0.0022	44173.72	-
	0.2250	44172.39	-
	0.2000	44172.34	-
	0.0400	44171.79	-
	0.2263	44171.44	-
	0.0400	44170.79	-
	0.0058	44170.27	-
	0.2260	44168.81	-
	0.0400	44168.73	-
	0.1000	44167.63	-
	0.2000	44167.25	-
	0.8059	44165.37	-
	0.0500	44165.31	-
	0.0066	44164.60	-
	0.0300	44164.24	-
	0.0315	44163.90	-
	0.6821	44162.59	-
	0.6325	44162.58	-
	0.6107	44162.14	-
	0.2000	44161.52	-
	0.0006	44160.86	-
Aggregation		0.01	+

Bonus: make the widgets configurable by the user